SenseCap: An Obstacle Awareness and Safety Headwear for the Visually Impaired

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Abstract

This paper presents the development of SenseCap, a smart wearable assistive cap designed to enhance safety and independent mobility for visually impaired users. The system integrates real-time environmental sensing with intelligent object detection to provide timely feedback on surrounding obstacles. SenseCap is implemented using a multi-layer mechatronic architecture, where a microcontroller unit (MCU) handles time-critical sensor data acquisition and feedback control, while a microprocessor unit (MPU) performs higher-level vision processing using an AI-based vision module. Ultrasonic sensing and inertial measurement data are fused to detect obstacles and head-level hazards, while haptic and audio feedback mechanisms convey intuitive alerts to the user. The system demonstrates effective task partitioning between hardware layers, ensuring reliable real-time performance. Experimental evaluation shows low response latency and consistent obstacle detection accuracy, validating SenseCap as a robust and practical wearable solution for improving navigation safety and situational awareness among visually impaired individuals.

1. Introduction

Safe and autonomous mobility is severely restricted by visual impairment, especially in settings with dynamic obstacles and head-level risks [1]. To navigate their environment, people with vision impairments frequently rely on traditional mobility aids like guiding dogs and white canes. These tools are useful for identifying obstacles at ground level and for basic orientation, but they are not very good for distinguishing moving objects, complex environmental conditions, or high hazards [2]. Because of this, visually impaired users are more likely to collide with low-hanging buildings, tree branches, and signboards, which could result in bodily harm and diminished confidence when navigating through crowded or unfamiliar areas.

Researchers have investigated the use of wearable assistive technology and electronic travel aids to improve visually impaired users' perception of their surroundings in order to overcome these restrictions [3]. Inertial measurement units (IMUs) allow head movement and orientation monitoring, while wearable devices with ultrasonic sensors can offer real-time distance measurements for obstacle detection. These sensor-based methods enhance situational awareness and enable prompt obstacle detection at various heights. However, these systems may have trouble in complicated situations where object recognition and classification are necessary, and they frequently offer little contextual information [4].

Through vision-based methods, recent developments in artificial intelligence and embedded systems have significantly increased the capabilities of assistive technologies. AI-driven vision modules provide deeper environmental information beyond basic distance measuring by enabling item detection and scene comprehension [5]. Assistive technology can more precisely detect threats and adjust to changing surroundings by fusing sensor data with vision-based perception. However, there are issues with computing burden, battery consumption, and response latency when integrating AI-based vision processing in wearable systems, especially when real-time input is needed.

Therefore, a small and effective wearable assistive system that successfully combines multi-sensor data and AI-based visual processing while preserving dependable real-time performance is still required. By utilising microcontroller units (MCUs) for real-time sensing and feedback and microprocessor units (MPUs) for sophisticated vision processing, a layered mechatronic architecture that divides time-sensitive activities from high-level processing might overcome these difficulties. This method lays the groundwork for creating useful wearable solutions that improve visually impaired people's situational awareness and navigation safety.

2. Methodology

2.1 Project Development

Overall, a block diagram is an overview of the entire system such that the connection between input, output and microcontroller can be explained. The project on SenseHat has 3 circuits which are categorized as system 1, system 2 and system 3. The block diagram of system 1 (emergency alert system), system 2 (obstacle detection system) and system 3 (object recognition system) is displayed in Fig. 1. The following diagram demonstrates the collaboration of the three mechanisms that secure the efficiency of the project in general. Fig. 2 indicates the flowchart of the system 1, system 2 and system 3 that represents the work principle of SenseHat project.

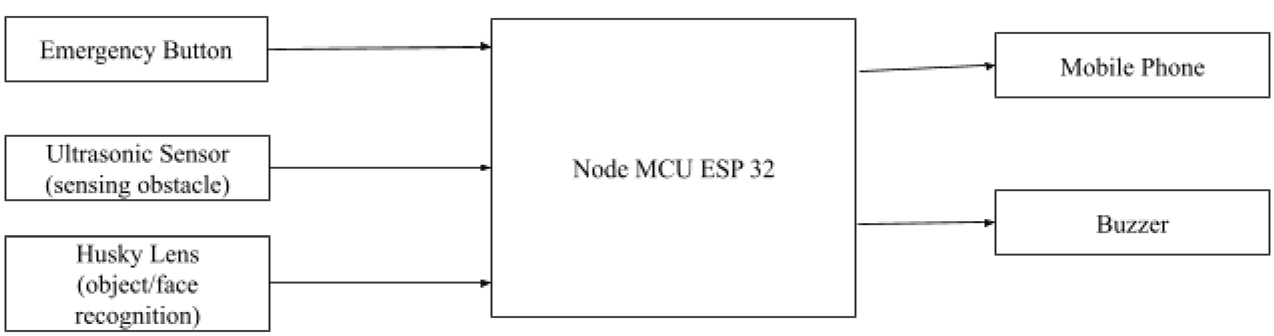

Fig.1 Block diagram of system 1 (emergency alert system), system 2 (obstacle detection system) and system 3 (object recognition system)

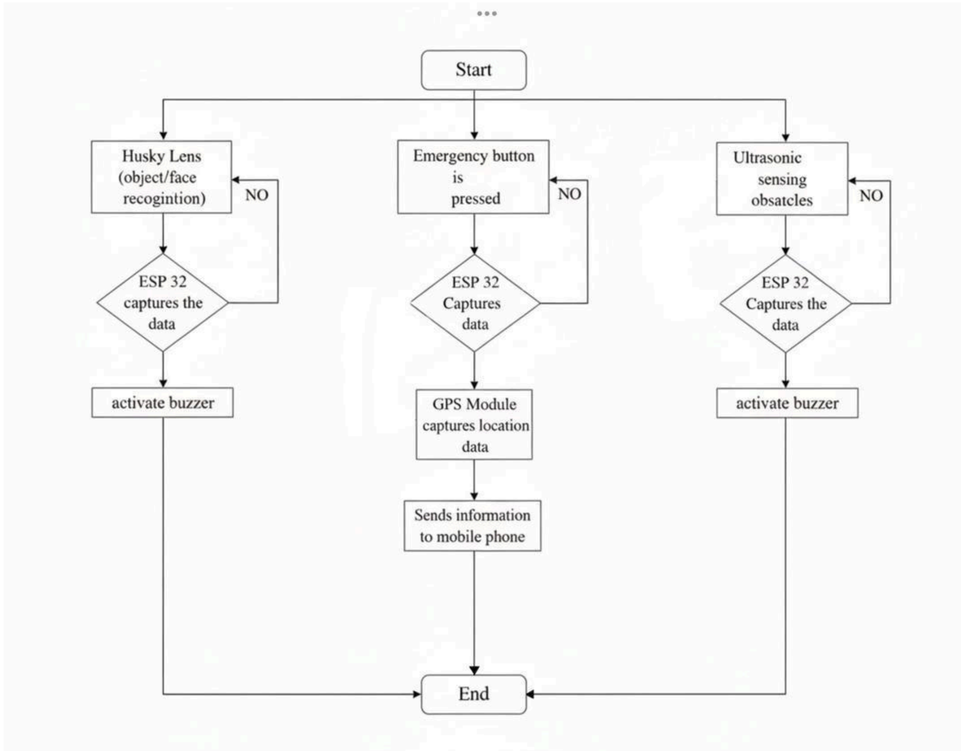

Fig. 2 The flow chart that explains all the work principles for the SenseHat project.

2.2 **Schematic Diagram**

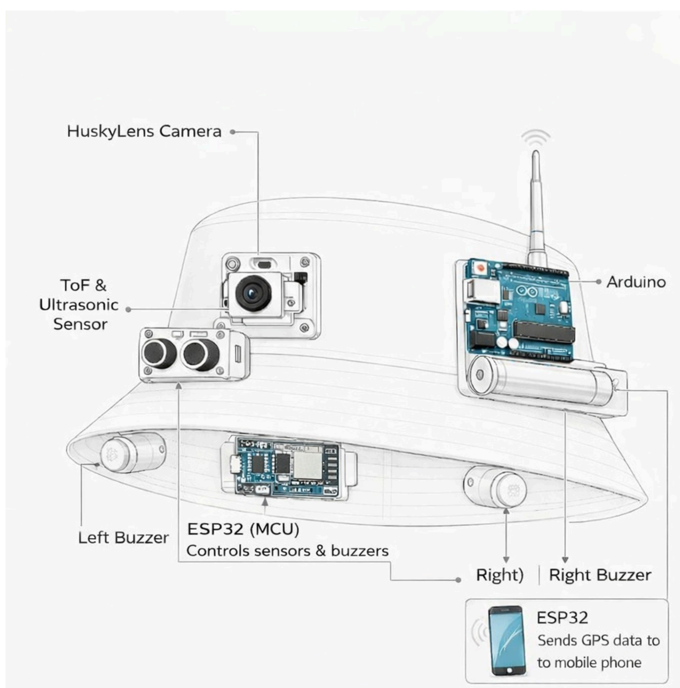

Fig. 3 The SenseCap